Chapter 13 — Halogens

Board-style MCQs · PECTAA Chemistry 11 · Answer key at end

Practice MCQs

1. Chlorine at room temperature is:

(a) Greenish-yellow gas **(b)** Red-brown liquid **(c)** Grey solid **(d)** Colourless liquid

2. Bromine at room temperature is:

(a) Red-brown liquid **(b)** Grey solid **(c)** Gas **(d)** Colourless gas

3. Iodine at room temperature is:

(a) Gas **(b)** White powder **(c)** Greyish-black solid **(d)** Liquid

4. Volatility from Cl_2 to I_2:

(a) Increases **(b)** Same **(c)** Zero **(d)** Decreases

5. London forces increase with:

(a) Lower mass **(b)** Smaller size **(c)** Electron number **(d)** Fewer electrons

6. Strongest oxidizing halogen:

(a) I_2 **(b)** F_2 **(c)** Cl_2 **(d)** Br_2

7. Chlorine displaces bromine from:

(a) KCl solution **(b)** Water only **(c)** KBr solution **(d)** NaF solution

8. F_2 bond is weak due to:

(a) Ionic character **(b)** Large size **(c)** No electrons **(d)** Lone-pair repulsion

9. H_2 + F_2 reaction is:

(a) Only on catalyst **(b)** Explosive **(c)** Very slow **(d)** No reaction

10. H_2 + Cl_2 needs:

(a) Darkness **(b)** Low pressure only **(c)** No initiator **(d)** Sunlight

11. Thermal stability of HX:

(a) All equal **(b)** HCl > HF > HI **(c)** HF > HCl > HBr > HI **(d)** HI > HBr > HCl > HF

12. Strongest reducing halide ion:

(a) Cl^- **(b)** Br^- **(c)** F^- **(d)** I^-

13. AgCl precipitate colour:

(a) White **(b)** Black **(c)** Yellow **(d)** Cream

14. AgBr precipitate colour:

(a) Brown **(b)** White **(c)** Cream **(d)** Yellow

15. AgI precipitate colour:

(a) Cream **(b)** Yellow **(c)** White **(d)** Green

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16. AgCl dissolves in:

(a) Concentrated HCl only **(b)** Water only **(c)** Dilute ammonia **(d)** NaCl solution

17. AgI is:

(a) Insoluble in ammonia **(b)** Soluble in water **(c)** Soluble in dilute ammonia **(d)** Colourless

18. NaCl + conc H_2SO_4 gives:

(a) SO_2 **(b)** Cl_2 **(c)** H_2S **(d)** HCl

19. NaBr + conc H_2SO_4 gives:

(a) No reaction **(b)** Only H_2 **(c)** Br_2 and SO_2 **(d)** Only HBr

20. NaI + conc H_2SO_4 can give:

(a) Only H_2 **(b)** I_2 and H_2S **(c)** No reaction **(d)** Only HI

21. Cl_2 + cold dilute NaOH gives:

(a) $NaClO_3$ **(b)** NaCl only **(c)** NaClO **(d)** Na_2O

22. Cl_2 + hot conc NaOH gives:

(a) NaOH only **(b)** NaClO **(c)** $NaClO_3$ **(d)** NaCl only

23. Chlorine in water forms:

(a) NaCl **(b)** HOCl **(c)** O_2 only **(d)** $HClO_3$

24. Main disinfecting species:

(a) HOCl **(b)** O_2 **(c)** Cl^- **(d)** Na^+

25. Lower pH favours:

(a) $NaClO_3$ **(b)** ClO^- **(c)** HOCl **(d)** HClO4

26. Residual chlorine provides:

(a) Continued protection **(b)** Taste only **(c)** Colour only **(d)** No effect

27. Halogens belong to group:

(a) 2 **(b)** 17 **(c)** 1 **(d)** 18

28. Halogen molecules are:

(a) Monatomic **(b)** Ionic **(c)** Diatomic **(d)** Triatomic

29. Oxidizing power down group:

(a) Same **(b)** Decreases **(c)** Zero **(d)** Increases

30. Iodine vapour colour:

(a) Yellow **(b)** Brown **(c)** Violet **(d)** Green

31. Halogens belong to:

(a) Group 18 **(b)** Group 1 **(c)** Group 2 **(d)** Group 17

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32. Chlorine at room temperature is:

(a) Colourless liquid **(b)** Red-brown liquid **(c)** Grey solid **(d)** Greenish-yellow gas

33. Bromine at room temperature is:

(a) Red-brown liquid **(b)** Grey solid **(c)** Colourless gas **(d)** Gas

34. Iodine at room temperature is:

(a) Greyish-black solid **(b)** Gas **(c)** White powder **(d)** Liquid

35. Volatility from Cl_2 to I_2:

(a) Increases **(b)** Decreases **(c)** Same **(d)** Zero

36. London forces increase with:

(a) Less polarizability **(b)** Higher temperature only **(c)** Molecular size **(d)** Lower mass

37. Bond strength generally decreases from:

(a) F_2 to Cl_2 only **(b)** I_2 to Cl_2 **(c)** Cl_2 to I_2 **(d)** No trend

38. F_2 bond is weak due to:

(a) Lone pair repulsion **(b)** Metallic bonding **(c)** Ionic character **(d)** Large size

39. Strongest oxidizing halogen is:

(a) I_2 **(b)** Cl_2 **(c)** Br_2 **(d)** F_2

40. Chlorine displaces bromine from:

(a) Water only **(b)** NaF solution **(c)** KBr solution **(d)** KCl solution

41. H_2 + Cl_2 reaction needs:

(a) Sunlight **(b)** No initiator **(c)** Darkness **(d)** Low pressure only

42. Thermal stability of HX decreases from:

(a) HI to HF **(b)** HF to HI **(c)** HCl to HF **(d)** No trend

43. Strongest reducing halide ion is:

(a) Br^- **(b)** I^- **(c)** Cl^- **(d)** F^-

44. AgCl precipitate is:

(a) Yellow **(b)** White **(c)** Cream **(d)** Black

45. AgBr precipitate is:

(a) Cream **(b)** White **(c)** Yellow **(d)** Black

46. AgI precipitate is:

(a) Cream **(b)** Black **(c)** White **(d)** Yellow

47. NaCl with conc H_2SO_4 gives:

(a) HCl **(b)** Cl_2 **(c)** H_2S **(d)** SO_2

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48. NaBr with conc H_2SO_4 gives:

(a) H_2S only **(b)** Br_2 and SO_2 **(c)** No reaction **(d)** HBr only

49. Cold dilute NaOH with Cl_2 gives:

(a) $NaClO_3$ only **(b)** NaCl and $NaClO_3$ **(c)** NaCl and NaClO **(d)** NaCl only

50. Main disinfecting species in chlorinated water:

(a) Cl^- **(b)** HOCl **(c)** Na^+ **(d)** O_2

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Answer Key with Brief Reasons

1. Answer: (a) Greenish-yellow gas — Halogen state.

2. Answer: (a) Red-brown liquid — Halogen state.

3. Answer: (c) Greyish-black solid — Halogen state.

4. Answer: (d) Decreases — Stronger London forces.

5. Answer: (c) Electron number — Polarizability.

6. Answer: (b) F_2 — Highest tendency to gain e^-.

7. Answer: (c) KBr solution — More reactive halogen.

8. Answer: (d) Lone-pair repulsion — Small F atoms.

9. Answer: (b) Explosive — High reactivity.

10. Answer: (d) Sunlight — Photochemical.

11. Answer: (c) HF > HCl > HBr > HI — Bond strength trend.

12. Answer: (d) I^- — Loses e^- easily.

13. Answer: (a) White — Halide test.

14. Answer: (c) Cream — Halide test.

15. Answer: (b) Yellow — Halide test.

16. Answer: (c) Dilute ammonia — Complex formation.

17. Answer: (a) Insoluble in ammonia — Halide test.

18. Answer: (d) HCl — No oxidation of Cl^-.

19. Answer: (c) Br_2 and SO_2 — Br^- oxidized.

20. Answer: (b) I_2 and H_2S — I^- strong reducer.

21. Answer: (c) NaClO — Disproportionation.

22. Answer: (c) $NaClO_3$ — Disproportionation.

23. Answer: (b) HOCl — Disinfection.

24. Answer: (a) HOCl — Oxidizing agent.

25. Answer: (c) HOCl — More effective disinfection.

26. Answer: (a) Continued protection — Water supply safety.

27. Answer: (b) 17 — VIIA.

28. Answer: (c) Diatomic — X_2.

29. Answer: (b) Decreases — Atomic size increases.

30. Answer: (c) Violet — I_2 vapour.

31. Answer: (d) Group 17 — Valence ns2 np5.

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32. Answer: (d) Greenish-yellow gas — Physical state.

33. Answer: (a) Red-brown liquid — Physical state.

34. Answer: (a) Greyish-black solid — Physical state.

35. Answer: (b) Decreases — Stronger London forces.

36. Answer: (c) Molecular size — Larger electron cloud.

37. Answer: (c) Cl_2 to I_2 — Longer bonds.

38. Answer: (a) Lone pair repulsion — Small atoms.

39. Answer: (d) F_2 — Greatest electron gain tendency.

40. Answer: (c) KBr solution — Stronger oxidant.

41. Answer: (a) Sunlight — Photochemical.

42. Answer: (b) HF to HI — Weaker H-X bond.

43. Answer: (b) I^- — Easiest to oxidize.

44. Answer: (b) White — Silver halide test.

45. Answer: (a) Cream — Silver halide test.

46. Answer: (d) Yellow — Silver halide test.

47. Answer: (a) HCl — No redox.

48. Answer: (b) Br_2 and SO_2 — Redox.

49. Answer: (c) NaCl and NaClO — Disproportionation.

50. Answer: (b) HOCl — Penetrates cells.

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